

OpticsTalk: Advanced ZPL macro programming

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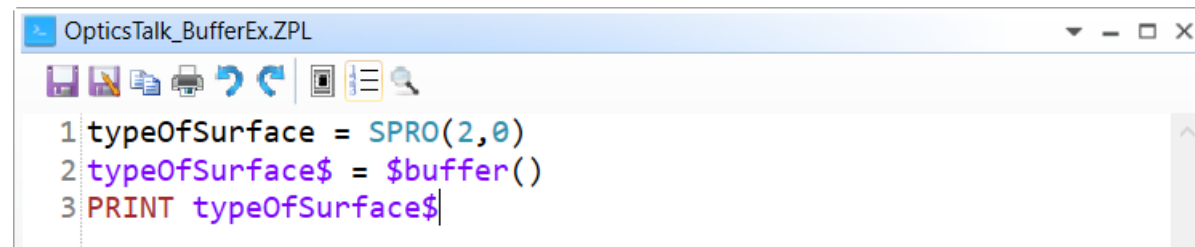
8/18/2020

Apply Make Thermal: Procedure

1. Enter number of configs, min, and max temperature.
2. Set up the format of the MC files to be opened.
 - a. Discussed in more detail: <https://my.zemax.com/en-US/Knowledge-Base/kb-article/?ka=KA-01485>
3. Open an MC file for editing.
4. Clear the Multi-Configuration Editor.
5. Insert new configurations based on the input *configNUM*
6. Apply each standard operand (Radius, Thickness, Clear Semi-Diameter, Chip Zone, Mechanical Semi-Diameter).
7. Apply the Thermal Pickup solves.
8. Update lens, and save output.

Can we expand the functionality?

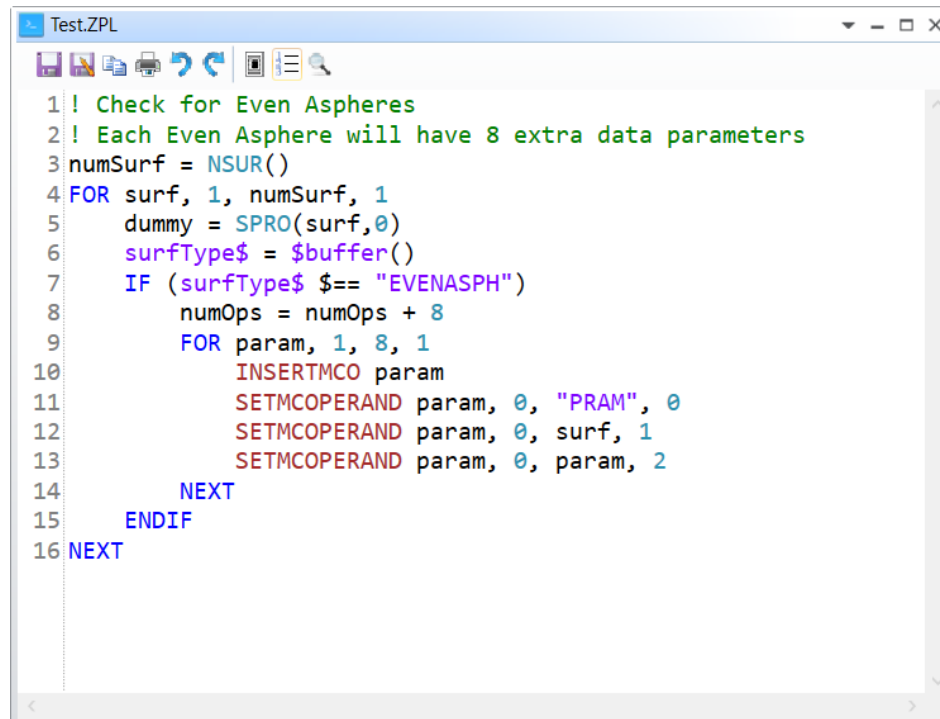
- The coefficients for Even Aspheres will have additional Multi-Configuration operands than the standard ones already added. We can check for the surface type with the BUFFER command.
- \$BUFFER():
 - Numeric Functions will return numeric values (marked as black in the ZPL)
 - Occasionally, certain settings for some Numeric Functions will return a string value. In that case, the string is placed in the "buffer". To extract it, use the String Function \$BUFFER()



```
OpticsTalk_BufferEx.ZPL
1 typeOfSurface = SPRO(2,0)
2 typeOfSurface$ = $buffer()
3 PRINT typeOfSurface$
```

Update the macro

- Add a Numeric Function which checks on surface properties. Use this Numeric Function to pull the Surface Type. Use \$BUFFER() to return the string.
- Add an IF statement to check for an Even Asphere. If one exists, add and edit more Multi-Configuration operands



```
1 ! Check for Even Aspheres
2 ! Each Even Asphere will have 8 extra data parameters
3 numSurf = NSUR()
4 FOR surf, 1, numSurf, 1
5     dummy = SPRO(surf,0)
6     surfType$ = $buffer()
7     IF (surfType$ == "EVENASPH")
8         numOps = numOps + 8
9         FOR param, 1, 8, 1
10             INSERTMCO param
11             SETMCOPERAND param, 0, "PRAM", 0
12             SETMCOPERAND param, 0, surf, 1
13             SETMCOPERAND param, 0, param, 2
14         NEXT
15     ENDIF
16 NEXT
```